FRESH STOCK

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GROUP NAME: OCHOBOT

1.0 Introduction

1.1 Background of case study

Food security is often viewed through the lens of large-scale agriculture, but it is fundamentally decided at the level of local consumption and business management. Food security literacy refers to the understanding of stakeholders, particularly Micro, Small, and Medium Enterprises (MSMEs), about the vulnerabilities in the food supply chain and how their operational choices affect it.

In the Association of Southeast Asian Nations (ASEAN) context, food waste in the retail and service sectors accounts for a significant portion of total landfill contributions. When a business overstocks and eventually throws away food, it doesn't just lose money; it wastes the water, energy, and labor used to produce that food, further straining a vulnerable ecosystem. Our project, FreshStock, serves as an educational and operational platform that bridges the gap between daily business tasks and broader food security goals. Through the combination of market simulations and waste tracking, we're helping business owners shift from a reactive approach to a more proactive, data-informed way of managing their operations.

1.2 Chosen SDG

a) SDG 2: Zero Hunger - By minimizing waste at the retail level, we ensure more food stays within the supply chain, contributing to overall food availability.
b) SDG 12: Responsible Consumption and Production - We directly target Target 12.3, halving per capita global food waste at the retail and consumer levels.

1.3 Problem Statement

Food businesses throughout ASEAN are experiencing a harder time managing their inventory since demand is shifting, ingredient prices are going up, and the supply chain is not always readily accessible. A lot of small and medium-sized businesses use manual estimation or past experience to figure out how much stock they have, which often leads to overstocking of perishable goods or shortages that come out of nowhere. Overstocking causes a lot of food waste and revenue loss, while shortages slow down operations while leaving customers less satisfied. Also, sudden price increases in key ingredients make profit margins even tighter because businesses don't have access to real-time data or alternative methods to get the ingredients they need. Food businesses have trouble running in a way that is good for the environment while making the food system more resilient without smart inventory systems and data-driven forecasting.

1.4 Objectives

a. To reduce food waste in businesses by analysing product stock, predicting weekly stock requirements, and preventing overstocking of perishable goods.
b. To improve demand forecasting with Artificial Intelligence (AI) by using historical data to forecast future demand, and adjusting predictions based on trends.
c. To optimise cost management by detecting price spikes, recommending cheaper or alternative meal options, and suggesting ingredient substitutions without lowering quality
d. To promote sustainable business practices by lowering carbon footprint from food waste, improving financial sustainability of MSMEs, and supporting ASEAN food security goals

2.0 Review of Existing Solutions

There are currently two main types of inventory and food waste management solutions, which are high-level enterprise software and platforms for post-production redistribution. These solutions work well in their own areas, but they often fail to meet the specific needs of MSMEs when it comes to Food Security Literacy and taking steps to avoid risks.

2.1 Conventional Inventory and POS Systems

Standard point-of-sale (POS) systems are great at keeping track of past sales and stock levels in real time. However, these platforms are mostly for transactions and only able to make limited predictions, failing to take into account market or environment changes. For most business owners, these systems are more to digital ledgers than strategic advisors. This implies that the user has to do all the work to prevent waste from happening.

2.2 Reactive Solutions and Food Redistribution

Most systems let businesses sell extra food at a lower price. These apps help cut down on waste, but they are "reactive" solutions. They only deal with food waste after it has been made and unsold. From a food security viewpoint, “proactive” solutions are more effective to prevent the initial overproduction of perishable goods than to rely on redistribution.

2.3 Enterprise Resource Planning (ERP) Software

Advanced ERP systems, like SAP's Supply Chain Management, use advanced AI to predict demand and improve logistics. These systems are better in terms of technology, but they are made for big companies that run business in more than one country. Their high costs and steep learning curves make them hard for most ASEAN small food business owners or households to use. They need a simpler, more intuitive interface.

2.4 The Identified Gap: Proactive Literacy and Simulation

There is a clear lack of a "proactive middle" solution that brings together operational management and strategic education. Existing solutions do not offer accessible AI predictions for people who are unfamiliar with data science. They also do not include risk simulation for users to see how external factors affect their revenue, and educational feedback that provide literacy for users. FreshStock fills this gap by combining a Situation Solver and a Waste Tracker into a single dashboard. This makes managing inventory less of a chore and more of a tool for making food more resilient in the region.

3.0 Methodology

The methodology that has been used for the development and operational logic of our project, FreshStock, is a combination of sustainable food management theories and data-driven system architectures. In order for us to tackle the real world issues and identify the gaps, the methodology has been divided into different conceptual frameworks and technical approaches.

3.1 Conceptual Framework : Food Waste Hierarchy

The food waste hierarchy methodology is one of the core logic that has been implemented into our project mainly because this framework is widely used in the environmental science and food security literature. This framework divides into two different priorities where the first one being Prevention or known as the top priority. This has been implemented into our Stock Prediction and Sales Tracker where the system uses historical sales data to forecast the precise inventory needs of a certain area which directly supports the prevention of overproduction and over-purchasing. This is implemented based on research that indicates source reduction is the most effective method in order for us to minimize the environmental and economic impact of food systems.

The second priority is Reuse and Recycle where we find a method to also identify the gaps that cannot be solved. While it is undeniable that waste is unavoidable, the implementation of AI into this project helps generate context-specific suggestions based on the logged reason. For example, it provides solutions for unsold and expired items. This ensures that MSME is able to contribute to a circular economy rather than an immediate landfill disposal.

3.2 Predictive Analytics and Inventory Management

Based on our research on inventory management, it is identified that conventional methods of inventory management relies heavily on manual work which leads to sometimes having unprone errors leading to massive overstocking or stockouts. The method that has been introduced in this project is the Data-Drive Forecasting, where it utilizes a time-series mathematical approach to analyze the average daily sales in order to project the future demand of the products that are sold or harvested.

Besides that, a Margin Impact Analysis has also been implemented which automatically calculates the percentage increases and margin impacts using the formula (Margin = newCost - oldCost). This functionality is able to empower business owners to make micro-adjustments to menu pricing dynamically which is something very strategic to increase SME financial resilience during inflation.

3.3 System Architecture and AI Integration

The third methodology that has been applied aims to bridge the gap between complex enterprise software (ERP) and standard SME capabilities where we integrated a three-tier operational methodology.

a. **Data Ingestion Layer** : Users are able to input daily operational metrics such as sales, and waste quantities. The system is designed with a low-friction UI to encourage consistent data entry from clients and suppliers.
b. **Processing and AI Layer** : The system is able to calculate the Key Performance Indicators (KPI) such as the waste rate and profit loss. There is also an AI engine that scans input keywords in order to generate market simulations and explain macro-economic factors that have been affecting the supply chain.
c. **Actionable Output Layer** : Raw data is converted into visual dashboards and a gamified Food Security Score. This score is able to serve as an educational feedback loop, translating abstract data into a clear indicator of operational sustainability.

4.0 Software Functionalities

4.1 Dashboard

The **Dashboard** acts as the central command centre, aggregating data from all modules into a single view. Its primary output is the "Food Security Score", a composite 0–100 metric derived from the business's waste rate, stock levels, and sales activity. The score defaults to 100 and degrades as problems are detected, giving operators an intuitive signal of whether their inventory practices are contributing to or undermining local food security.

4.2 Product Stock

The **Product Stock** module maintains the inventory catalogue and serves as the data source for all other modules. Operators define products with stock quantities, selling prices, and optional ingredient breakdowns. This ingredient data enables automatic cost-of-goods calculation, which flows into profit margin displays and the Waste Tracker's cost price estimation.

4.3 Stock Prediction

The **Stock Prediction** module uses sales history from the Sales Tracker to forecast 7-day demand per product, comparing projected need against current stock to flag items at risk of stockout. Operators can also run predictions manually by inputting their own average daily sales estimate, which is useful when anticipated demand differs from historical patterns such as ahead of a public holiday or promotional period. The purpose is to shift restocking decisions from intuition-based to evidence-based, reducing both stockouts and over-purchasing.

4.4 Waste & Impact Tracker

The **Waste & Impact Tracker** is the system's primary AI-integrated feature. When an operator logs a waste event, the system calculates profit loss and waste rate automatically, then calls the *Google Gemini API* to generate a contextual recommendation based on the product type and waste reason. The purpose is to close the gap between waste occurring and corrective action being taken, rather than operators discovering losses after the fact, the system surfaces actionable guidance at the point of logging.

4.5 Market Price Intelligence

The **Market Price Intelligence** module addresses supplier-side cost pressure. Operators log ingredient price changes, and the system instantly calculates margin impact and recommends a selling price adjustment to protect profitability. The intent is to give small operators the same cost visibility that larger businesses achieve through dedicated procurement systems.

4.6 Sales Tracker

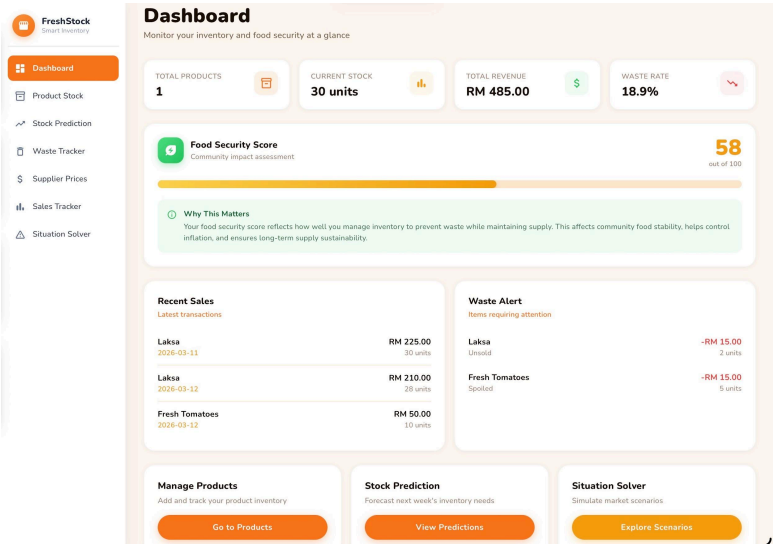
The **Sales Tracker** functions as the transactional backbone. Recorded sales entries are aggregated into revenue trends, product performance rankings, and summary KPIs. Critically, this data feeds the **Stock Prediction** module, which uses historical average daily sales to forecast 7-day demand per product and flag items at risk of stockout. This turns reactive restocking, typically done by intuition, into a data-driven process.

4.7 Situation Problem Solver

The **Situation Problem Solver** takes a forward-looking approach by simulating the impact of real-world disruptions, floods, fuel price surges, export bans, port delays, on pricing and waste before they occur. Rather than responding to crises reactively, operators can use the module to stress-test their inventory strategy and prepare contingency responses in advance.

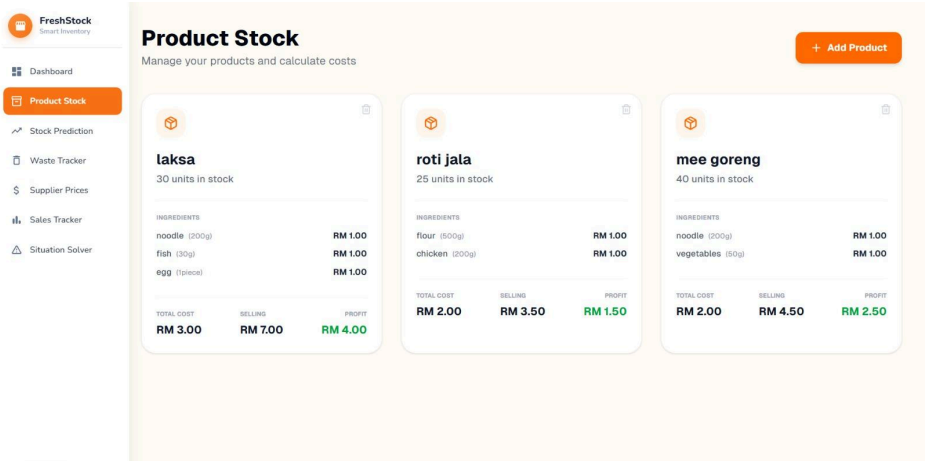
5.0 Prototype Showcase

5.1 Dashboard

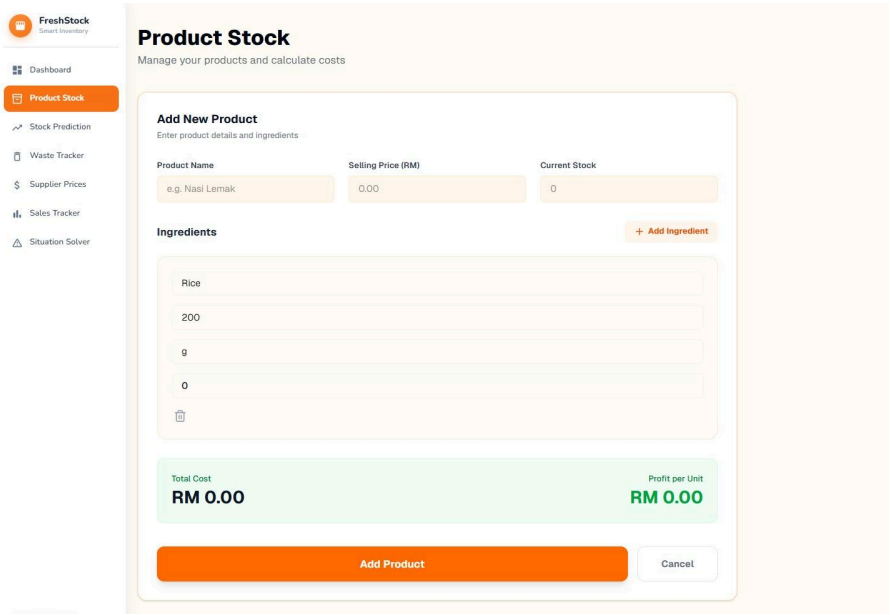


Display the total products, current stock, total revenue, and waste rate based on updated data. Also, featuring food security scores to acknowledge users on how good they are in managing food waste.

5.2 Product stock

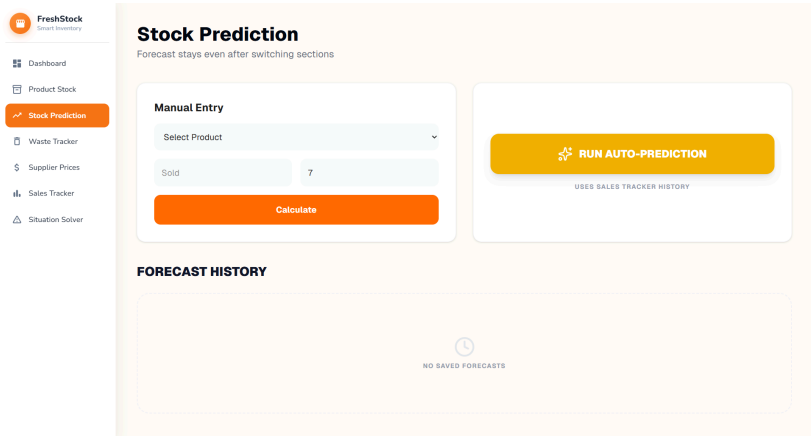


Display all products added by users including ingredients needed for each product with calculated cost and profit.



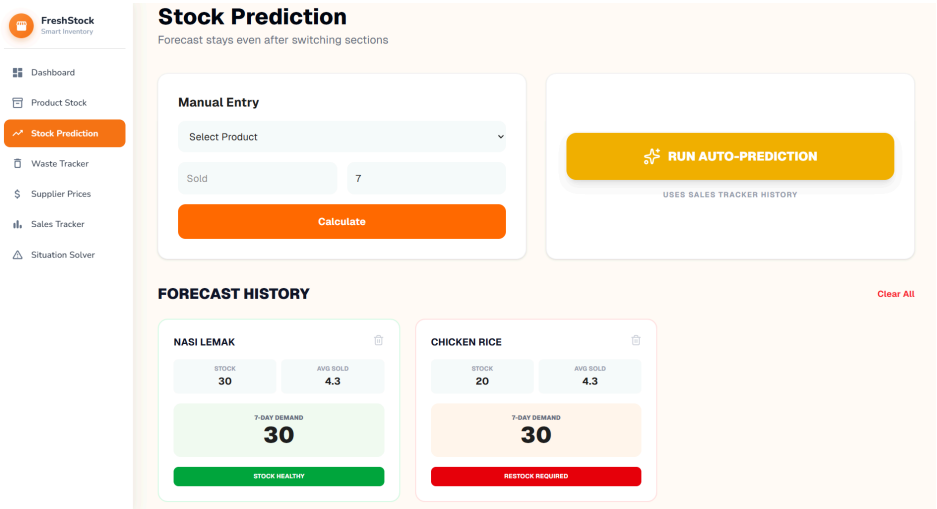
Once users click the 'Add Product' button, they are required to fill in all the details of the product to determine total cost and profit.

5.3 Stock Prediction



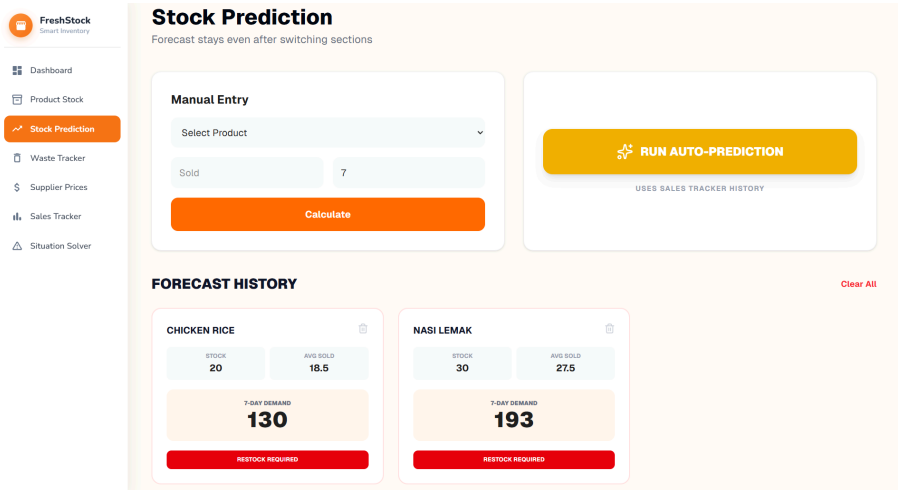
The Stock Prediction section provides two ways of predicting; manual prediction and auto prediction.

5.3.1 Manual Prediction



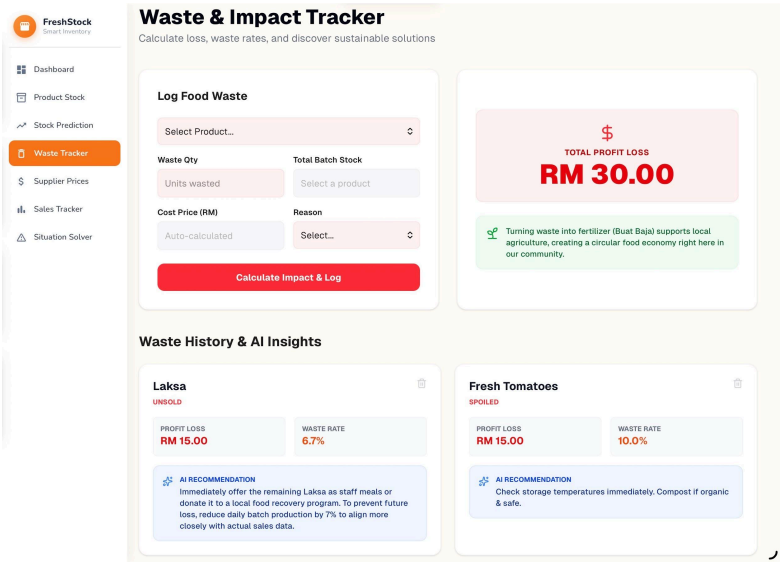
After the user fills in all details, the right box shows the result if the current stock is not sufficient for next week's sale. The left box shows the result if the current stock is enough for next week's sale.

5.3.2 Auto Prediction



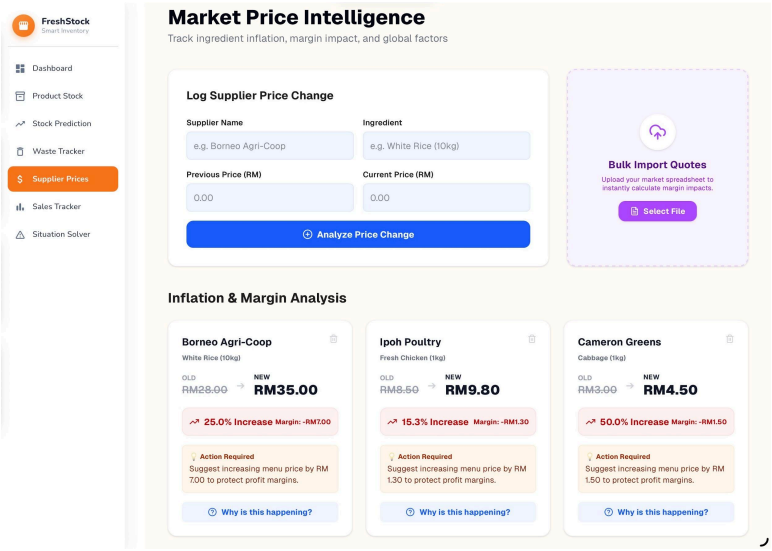
Auto prediction uses data received from the sale recorded in the Sales Tracker section. It will automatically generate analysis on stock required based on the trend and demand.

5.4 Waste Tracker



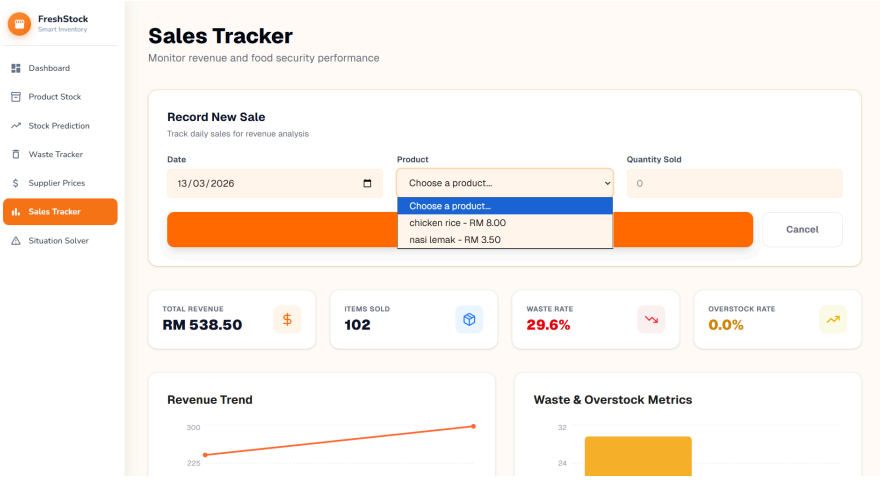
When users log food waste, it will automatically calculate the profit loss and waste rate with AI recommendations on how to manage the waste.

5.5 Supplier Prices

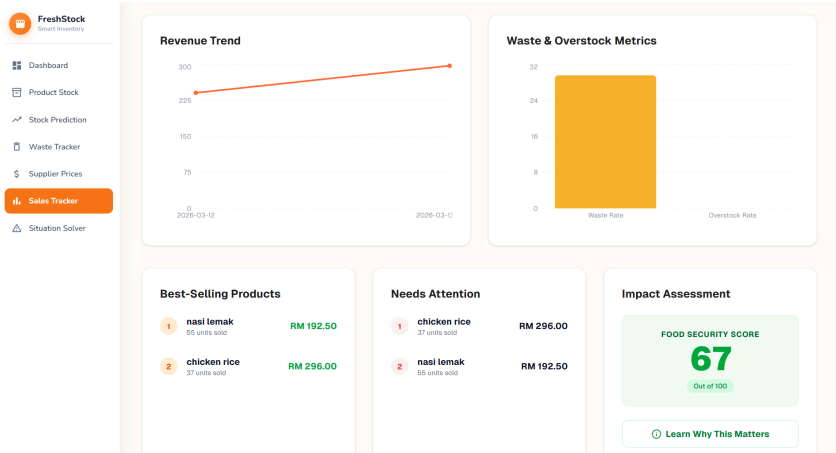


The Supplier Price section provides inflation and margin analysis which helps users to update their current price according to the changes in market price based on the suggestion given by AI. It also explains the reasons behind the price changes to users to improve their understanding in business management.

5.6 Sales Tracker

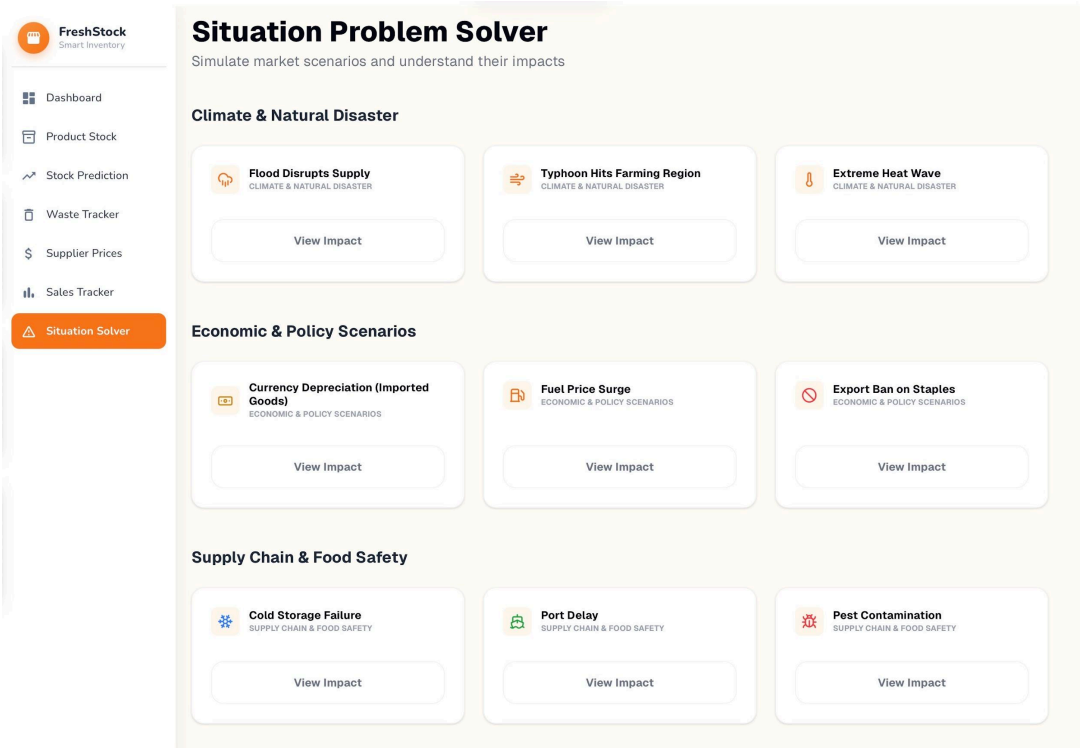


Users are required to record sales for each product for every day to see the revenue trend and rate of waste and over-stock.



Graphs will be generated, best and worst selling products will be listed with food security scores.

5.7 Situation Solver



Situation Problem Solver display multiple scenarios with different possible impacts to educate users about food security and help them to be prepared if it happens in real life.

6.0 Result

The primary objective of this project is to bridge the gap between retail operations especially in the food and beverages industry and broader community food stability. The results of the development phase demonstrate that FreshStock website successfully operates as a fully integrated and functional ecosystem. This project leverage decoupled React architecture, predictive mathematical modeling, and AI-driven insights. The website is able to translate raw inventory and sales data into actionable, educational metrics for vendors.

- **Ecosystem Integration**: Real-time synchronization of data across Sales Tracker, Stock Prediction, Product Stock and Waste Tracker modules.
- **The 'Food Security Score'**: Validation of algorithm which quantifies retailers impact.
- **Predictive Shortage Prevention**: The system is able to forecast inventory needs and trigger restock alerts automatically.
- **AI-Driven Waste Mitigation**: The generation of localized, actionable and strategies to rescue food that might go to waste.
- **Situational Problem Solver**: The validation of possible real-world scenarios revolving macro-economic crises and possible mitigation strategies for vendors.

6.1 Ecosystem Integration

Result: FreshStock successfully achieved real-time data synchronization across independent modules without relying on a heavy backend database.

Mechanism: FreshStock mechanism utilizes optimized browser storage (localStorage) as centralized state manager. When a product is created or uploaded in the Product Stock module, the data is accessible across the entire website ecosystem.

Validation: By logging a sale in the Sales Tracker, it accurately fetches the latest product data and integrates waste data to perform cross-module metric calculations seamlessly.

6.2 The 'Food Security Score'

Result: FreshStock successfully aggregated disparate financial metrics into 'Food Security Score' to educate vendors on their localized impact.

Mechanism: The scoring algorithm averages three distinct pillars; the **wasteScore** which penalizes high waste rates, the **stockScore** which rewards consistency in inventory levels above 10 units, and the **salesScore** which measures market movement.

Validation: The module accurately calculated a live **overstockRate** by identifying items where current stock exceeded twice the average daily sales volume. This serves as an early warning for potential future food waste.

6.3 Predictive Shortage Prevention

Result: The Stock Prediction engine manages to forecast future supply requirements by analyzing recent sales histories.

Mechanism: The Stock Prediction module calculates average daily sales for specific items and projects the required inventory for upcoming week.

Validation: During testing, the module successfully triggered automated “Restock Required” alerts the moment projected weekly demand exceeded current inventory, preventing potential supply chain breaks.

6.4 AI-Driven Waste Mitigation

Result: The Waste Tracker calculates financial loss and integrates generative AI to actively rescue food at risk of waste.

Mechanism: When the vendor logs spoiled or unsold items, Waste Tracker calculates the exact waste rate and feeds the context to an AI API.

Validation: During testing the Waste Tracker engine returns localized, actionable advice such as using waste as fertilizer for community gardens. This educates vendors on circular economic principles.

6.5 Situational Problem Solver

Result: The Situation Solver engine able to stress-test vendor’s live inventory against macro-economic crises.

Mechanism: The Situational Solver engine applies specific mathematical modifiers (priceMod and wasteMod) mapped to specific events. For example, in a Flood scenario, a 15% logistic price increase and a 20% spoilage rate directly to the vendor’s live-stock count.

Validation: The Situational Solver successfully generates real-time revenueImpact and stockShortage projections, by assigning standardized riskLevel (Low to Critical). This engine

also triggers literacy alerts such as explaining how currency depreciation impacts the cost of imported dairy.

7.0 AI Acknowledgement

We acknowledge the use of Google's *Gemini 2.5 Flash* (https://gemini.google.com/), OpenAI's *GPT-5.3* (https://chat.openai.com/), and Anthropic's *Claude Sonnet 4.6* (https://claude.ai/) to help us improve our code and debug errors to reduce our time efficiently. Here are the prompt entered:

1. If I make a prediction for the first product, then I make a prediction for the second product, the prediction for the first product gets erased. can it just stay so the prediction area will have 2 products. The same goes if I make 3 or more product predictions.
2. How to fix this error? You're importing a component that needs `useEffect`. This React Hook only works in a Client Component. To fix, mark the file (or its parent) with the `"use client"` directive.
3. Is there anything wrong here? "dependencies": { "@emotion/react": "^11.14.0", "@emotion/styled": "^11.14.1", "@mui/icons-material": "^7.3.9", "@mui/material": "^7.3.9", "lucide-react": "^0.577.0", "next": "16.1.6", "react": "19.2.3", "react-dom": "19.2.3", "recharts": "^2.12.0" },
4. Can you check if there's any possible run-time error and any improvement i can do : (my code)
5. The waste graph is not fetching data for interpolation, identify the issue.
6. Suggest potential future improvements for our project.

The original output was adapted and modified for the final response.

8.0 Conclusion

This project is a complete inventory management system in a website structure which helps business owners to analyse their stocks while minimizing food waste efficiently. This website consolidates multiple functionalities such as Product Stock Management, Stock Prediction, Waste tracker, Supplier Prices Tracker, Sales Tracker and Situation Solver for market simulations. All these functionalities help users to monitor their inventory easily by forecasting available demand and analyze business performance according to current market conditions.

In the Product Stock Management section, it allows users to calculate the cost of each ingredient in every product and identify the profit acquired from the price allocated by users. The stock prediction provides two options, where users can pick either to use manual or automated projection which give immediate analysis about stock requirements. For Waste Tracker, it elevates sustainability by letting users key-in items that are not sold, being returned or already expired while providing solutions on how to lessen business's losses with the help of AI. Next, for Supplier Price Tracker, it helps in monitoring ingredients costs in case of increment or decrement of the current price. With the changes of ingredients costs, it assists business owners on suggesting price adjustments in order to maintain profit margins. Additionally, Sales Tracker monitors revenue and product performance by showing the graph of the trend and informing the best-selling and low-performing products and at the same time stating the food security score. The score can let users see if they achieve the objectives of minimizing food waste or not. Finally, we have a Situation Problem Solver where it simulates potential market scenarios and guides users in understanding the impacts of those particular scenarios to help them plan and be prepared.

Some refinements can be made to improve this project such as implementing real-time market and supplier APIs that can automatically update changes in prices for ingredients without the users having to update it manually. Moreover, the Product Stock section can integrate barcode scanning to update ingredients list in order to reduce manual input errors. Last but not least, by using the power of AI, the system may automatically give suggestions on where to donate or partnerships with food banks which not only support sustainability but also give a positive community impact.